

SSURGO Soil Analysis Report

Field Overview

- **Location:** Iowa, Grundy County
- **Coordinates:** 41.921630, -93.776263 to 41.914425, -93.771320
- **Field Size:** ~81 acres
- **Data Source:** USDA NRCS Soil Data Access (SDA)

Summary

This report presents the analysis of SSURGO (Soil Survey Geographic Database) soil data for an 80-acre agricultural field in Iowa. The analysis includes soil property maps, horizon-level data, and component-weighted averages for key soil characteristics.

Spatial Visualization

Figure 1: 3x3 panel showing: - Satellite basemap: Esri World Imagery - Soil polygons: Clipped to field boundary with 50% transparency - Classification: Natural Breaks (Jenks) with 3 classes - Polygon outlines: Green (#228B22) at 0.8px - Field boundary: Black outline at 2.5px - Headland buffer (30m): Red dashed outline 1. Field boundary (no soil data) 2. SSURGO soil map units 3. Organic Matter (%) 4. Soil pH 5. Available Water Capacity (inches/inch) 6. Clay percentage (%) 7. Sand percentage (%) 8. Headland buffer (30m) with OM overlay 9. CEC7 (meq/100g)

Map Unit Summary

Map Unit Key	Soil Name	Symbol	OM (%)	pH	AWC (in/in)	Clay (%)	Sand (%)	CEC7
2550272	Spillville loam, 2 to 5 percent slopes	485B	1.84	6.82	0.184	22.7	42.7	18.7
2550277	Blue Earth mucky silt loam, 0 to 1 perce	511	13.18	7.65	0.210	25.2	12.5	24.0
2550303	Zenor sandy loam, 2 to 5 percent slopes	828B	0.78	7.56	0.072	11.1	61.2	12.6
2550305	Zenor sandy loam, 5 to 9 percent slopes,	828C2	0.68	7.56	0.072	11.1	61.2	12.6
2765522	Clarion loam, Bemis moraine, 2 to 6 perc	L138B	0.88	7.38	0.183	17.7	46.9	13.7

Map Unit Key	Soil Name	Symbol	OM (%)	pH	AWC (in/in)	Clay (%)	Sand (%)	CEC7
2834849	Nicollet loam, 1 to 3 percent slopes	L55	1.33	7.40	0.185	19.1	43.4	14.8
2835012	Webster clay loam, Bemis moraine, 0 to 2	L107	2.06	7.50	0.181	22.4	38.1	17.7
2835146	Harps clay loam, Bemis moraine, 0 to 2 p	L95	2.06	7.91	0.185	20.9	39.6	14.4
2922007	Canisteo clay loam, Bemis moraine, 0 to	L507	1.93	7.83	0.180	22.1	39.5	16.2

Key Statistics

Organic Matter (%)

- Mean: 2.75
- Std Dev: 3.95
- Min: 0.68
- Max: 13.18

Soil pH

- Mean: 7.51
- Std Dev: 0.31
- Min: 6.82
- Max: 7.91

Available Water Capacity

- Mean: 0.16
- Std Dev: 0.05
- Min: 0.07
- Max: 0.21

Clay (%)

- Mean: 19.14
- Std Dev: 5.06
- Min: 11.06
- Max: 25.20

Sand (%)

- Mean: 42.79

**SSURGO Soil Analysis - Grundy County, Iowa (~81 acres)
Natural Breaks (Jenks) Classification - 3 Classes**

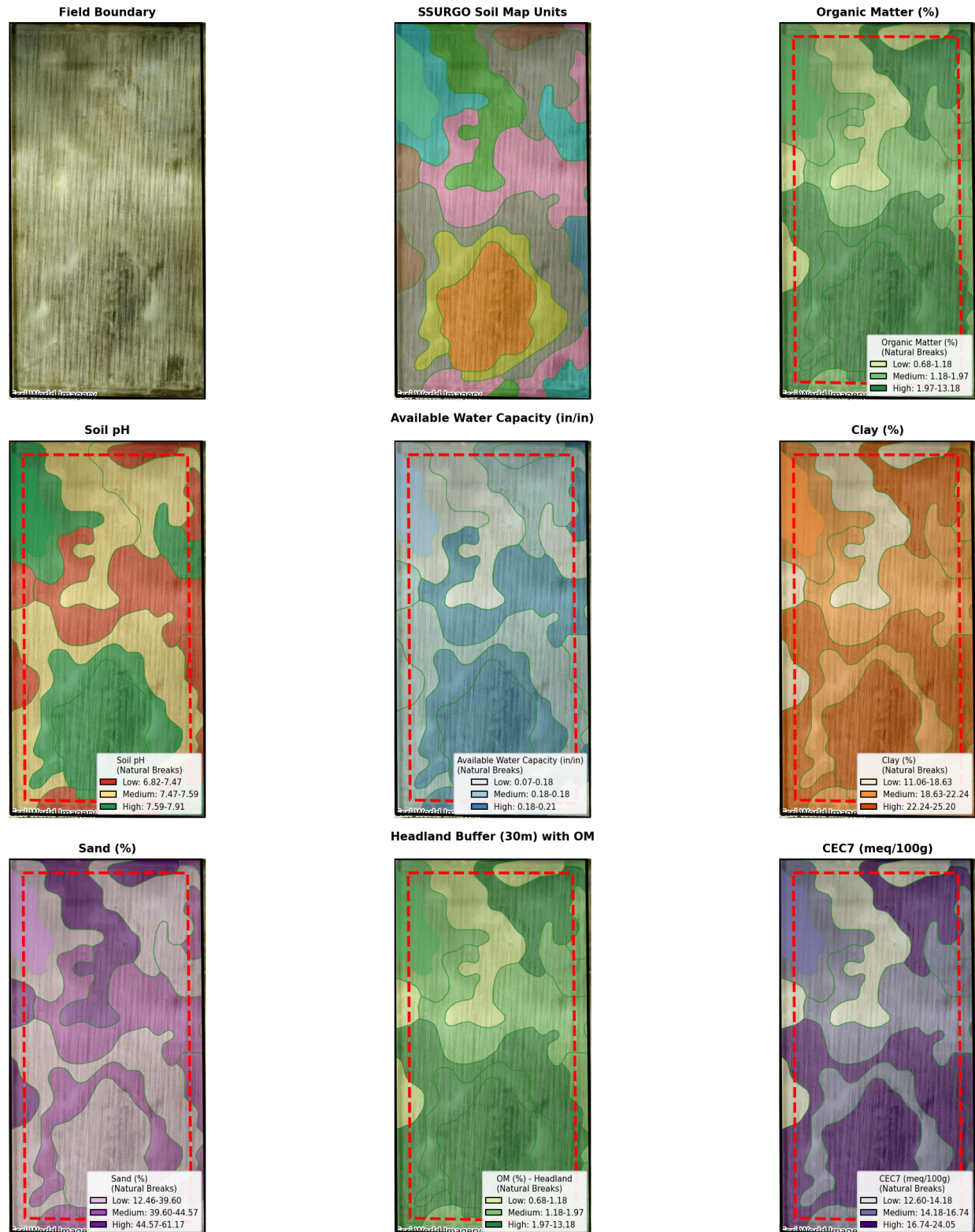


Figure 1: SSURGO Soil Maps

- Std Dev: 14.38
- Min: 12.46
- Max: 61.17

CEC7 (meq/100g)

- Mean: 16.10
- Std Dev: 3.67
- Min: 12.60
- Max: 24.05

Soil Properties Table

The following table shows detailed horizon-level data sorted by map unit, depth, and component percentage:

mukey	hzname	Depth (cm)	OM (%)	pH	AWC	Clay (%)	Sand (%)	CEC7	Comp %
2550272	Ap	0-23	3.50	6.7	0.210	24.0	40.0	20.5	90
2550272	Ap	0-20	6.00	6.7	0.210	31.0	22.0	26.4	10
2550272	A1,A2	20-81	4.50	6.7	0.210	31.0	19.0	26.1	10
2550272	A	23-112	2.50	6.7	0.180	24.0	40.0	20.3	90
2550272	AB	81-102	5.50	6.7	0.170	29.0	26.7	24.8	10
2550272	Bg1	102-112	1.50	6.7	0.150	22.0	30.0	18.4	10
2550272	C	112-200	0.25	7.0	0.180	20.0	50.0	15.7	90
2550272	Bg2	112-132	1.50	6.5	0.170	22.0	30.0	18.4	10
2550272	Cg	132-152	1.00	6.7	0.150	16.0	58.0	13.6	10
2550277	Lpco	0-25	17.50	7.6	0.220	25.0	7.0	26.1	85
2550277	Ap	0-23	7.00	7.5	0.180	32.0	30.0	25.0	10
2550277	Ap	0-20	7.00	7.5	0.180	32.0	30.0	23.5	4
2550277	Ap	0-25	7.00	7.8	0.210	15.0	50.0	12.1	1
2550277	Ak	20-41	5.00	8.0	0.180	32.0	32.0	23.2	4
2550277	A	23-41	5.00	7.7	0.180	32.0	32.0	25.5	10
2550277	Lco	25-173	17.50	7.6	0.220	25.0	7.0	26.1	85
2550277	A	25-46	5.00	7.8	0.140	15.0	55.0	12.0	1
2550277	AB	41-51	3.00	7.7	0.180	32.0	32.0	25.0	10
2550277	Bkg	41-160	1.50	8.0	0.180	25.0	39.0	16.9	4
2550277	Bg	46-74	1.50	7.8	0.130	10.0	65.0	7.9	1
2550277	Bkg	51-91	1.50	8.0	0.170	27.0	39.0	17.6	10
2550277	2Cg	74-200	0.25	8.0	0.060	5.0	90.0	2.8	1
2550277	Cg	91-200	0.25	8.0	0.180	21.0	44.0	15.0	10
2550277	Cg	160-200	0.25	8.0	0.180	21.0	44.0	14.3	4
2550277	2Cg	173-200	0.25	7.6	0.190	28.0	9.0	19.9	85
2550303	H1	0-30	2.00	6.5	0.110	12.5	67.9	17.5	95
2550303	H1	0-15	2.20	7.9	0.210	22.5	nan	17.5	5
2550303	H2	15-152	0.25	7.9	0.180	24.0	nan	12.5	5
2550303	H2	30-84	0.50	7.3	0.100	16.0	43.8	17.5	95
2550303	H3	84-152	0.50	8.2	0.020	5.0	79.2	6.5	95

Table continued: 105 total records

Methodology

Data Sources

- **USDA NRCS Soil Data Access (SDA):** <https://sdmdataaccess.sc.egov.usda.gov/>

- **SSURGO Database:** Soil Survey Geographic Database

Processing Steps

1. Field boundary created from GPS coordinates
2. Map unit keys (mukeys) extracted via SDA spatial intersection
3. Tabular data downloaded: mapunit, component, chorizon tables
4. Horizon properties aggregated using component-weighted averages
5. Natural Breaks (Jenks) classification applied for choropleth maps

Classification

- **Method:** Natural Breaks (Jenks)
- **Classes:** 3 per property
- **Map Projection:** EPSG:4326 (WGS84), UTM Zone 15N (EPSG:32615) for headland buffer

Findings

1. **Soil Diversity:** The field contains 9 distinct soil map units
2. **Dominant Soil Types:**
 - Clarion, Nicollet, Webster, and Harps loams (Bemis moraine)
 - Blue Earth mucky silt loam (poorly drained)
 - Spillville and Zenor loams (steeper slopes)
3. **Organic Matter Range:** 0.68% - 13.18%
 - Blue Earth soils have highest OM (13.18%) due to wetland origin
 - Zenor sandy loams have lowest OM (0.68%)
4. **pH Range:** 6.8 - 7.65
 - Generally neutral to slightly alkaline
 - Suitable for most crops
5. **Available Water Capacity:** 0.07 - 0.21 in/in
 - Blue Earth and Spillville have highest AWC
 - Zenor sandy loams have lowest AWC (drought-prone)

Data Files

File	Description
field_boundary.geojson	Field boundary polygon
headland_buffer.geojson	30m inward buffer for headland border
ssurgo_raw.geojson	Raw SSURGO polygons
ssurgo_enriched_correct.geojson	SSURGO polygons with corrected properties
ssurgo_field_polygons.geojson	SSURGO polygons clipped to field
soil_properties.csv	Mapunit-level aggregated properties
soil_summary_table.csv	Horizon-level detailed data
soil_maps_3x3.png	3x3 panel visualization with satellite basemap

Report generated from USDA NRCS SSURGO data